A Monitoring system for Greenhouse using LoRaWAN

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Abstract:

Exploring the integration of Internet of Things (IoT) technology, specifically utilizing a LoRaWAN network, to develop an environment monitoring system. The system aims to capture real-time data related to weather conditions, humidity, and other environmental factors, contributing to efficient and precise smart agriculture practices. The project emphasizes the use of wireless sensor networks and advanced sensing technologies to enhance the decision-making processes in farming. The proposed system not only facilitates automatic control of essential parameters such as watering and fertilization but also seeks to address the gap in technology adoption among traditional farmers. The research underscores the importance of leveraging innovative solutions to improve productivity, quality, and environmental sustainability in agriculture. The study emphasizes the significance of providing farmers with actionable insights derived from environmental measurements for informed decision-making, ultimately contributing to a more intelligent and sustainable farming ecosystem.

Keywords: LoRa, Internet of Things (IoT), Greenhouse, Monitor and Control.

1. Introduction

In an era where sustainable agriculture and climate consciousness are at the forefront of global concerns, the Lora Greenhouse Monitoring Project emerges as a beacon of innovation and environmental stewardship. This project sets out to revolutionize the way we cultivate crops, harnessing the power of LoRa (Long Range) technology to create smarter, more efficient, and eco-friendly greenhouse environments. Greenhouses play a vital role in meeting the growing demand for fresh, high-quality produce, year-round. However, optimizing the conditions within these controlled environments to ensure the health and productivity of crops is a complex endeavour. Traditional greenhouse management often relies on manual intervention and monitoring, which can be time-consuming, inefficient, and prone to human error. The Lora Greenhouse Monitoring Project addresses these challenges by integrating LoRa technology, a low-power, long-range wireless communication system. With this advanced system, we can remotely collect real-time data on temperature, humidity, soil moisture, light levels, and other critical environmental factors within the greenhouse. The data is transmitted wirelessly to a central control system, enabling precise, data-driven decision-making and automation of greenhouse operations. In this project, we will delve into the technology, methodology, and potential impact of LoRa greenhouse monitoring. We will explore the hardware and software

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components of the system, discuss the benefits it offers to growers, and consider the broader implications for sustainable agriculture. By harnessing the power of LoRa, the Lora Greenhouse Monitoring Project seeks to create a greener, more efficient, and data-informed future for greenhouse farming, aligning with the pressing need to make agriculture more resilient and environmentally responsible.

2. Purpose:

Exploring the implementation of an Internet of Things (IoT) based environment monitoring system using a LoRaWAN Network. The focus is on utilizing this system for real-time measurement and interpretation of environmental data, such as weather conditions and humidity, to facilitate efficient and precise agricultural practices. The goal is to integrate smart technologies into farming, providing farmers with valuable insights to enhance productivity and quality while maintaining environmentally friendly practices. The research aims to bridge the gap between traditional farming methods and modern smart agriculture solutions, addressing the specific needs of farmers in optimizing soil conditions, temperature, and gas levels for optimal crop quality and yield. The ultimate objective is to contribute to the advancement of intelligent farming systems that are accessible and beneficial to a broader range of agricultural practices.

3. Literature review

Sr No.	Title of paper	Year of paper publishing	Publisher	Methodology of paper
1	A Design of Greenhouse Monitoring & Control System Based on ZigBee Wireless Sensor Network	2007	IEEE Explore	Monitoring and controlling greenhouse with the help of ZigBee WSN.
2	LoRaWAN as Secondary Telemetry Communication System for Drone Delivery	2018	IEEE	Utilization of LoRaWAN as a secondary telemetry communication system for drone delivery which helps in communication.
3	LoRaWAN Network: Radio Propagation Models and Performance Evaluation in Various Environments in Lebanon	2019	IEEE Internet of Things	Analyses and categorizes existing research to identify challenges with LoRAWAN.

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4	Using LoRa Technology to Monitor and Control Sensors in the Greenhouse	2020	IOPSCIENCE	Monitoring and controlling sensors in greenhouse using LoRA Technology.

Table no.1 Literature review

4. Existing System approach

In the current scenario, traditional farming practices dominate the agricultural landscape. Farmers typically rely on manual methods for monitoring and managing their crops, lacking real-time data and precision in their agricultural tasks. The absence of advanced technologies like the Internet of Things (IoT) and Wireless Sensor Networks (WSN) limits the efficiency, accuracy, and economic benefits that can be achieved in farming.

5. Proposed-system Approach

Here we propose the development of an Environment Monitoring System utilizing a LoRaWAN Network for seamless integration with a web application. The primary goal is to enhance efficiency, accuracy, and economic benefits through real-time monitoring of environmental factors. The system approach for a monitoring system for greenhouse using WSN consist of 2 parts which are as follows 1.Transmitter Node 2.Receiving Node.

6. Proposed-System Block Diagram:

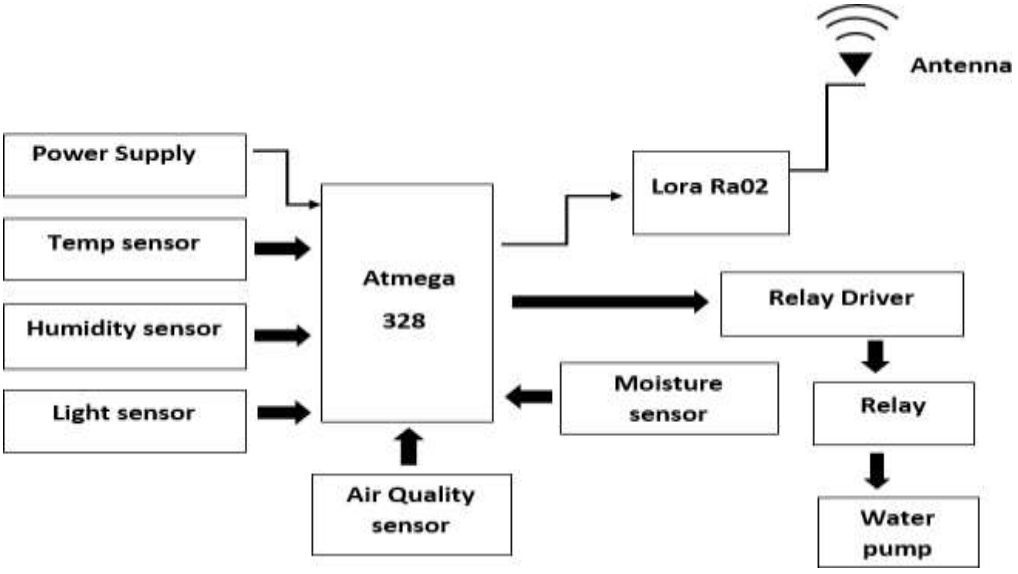

Figure no.1: Transmitting Node

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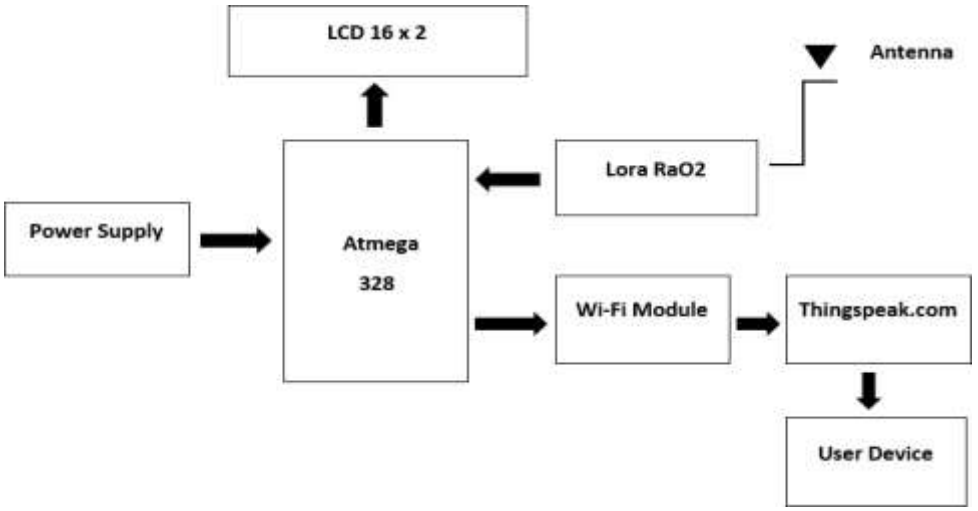

Figure no.2: Receiving Node

7. Block Diagram description

The transmitter side of the greenhouse monitoring system incorporates five essential sensors—temperature, humidity, moisture, light, and air quality sensors—to collect comprehensive data for monitoring the greenhouse environment. These sensors work in tandem to provide accurate readings that play a crucial role in predicting future yield growth details. The collected data is centralized at the Atmega 328 controller, which serves as the nerve center connecting to all sensors. Subsequently, a water pump, governed by a threshold logic, is integrated into the system. If the soil moisture levels surpass the defined threshold, the pump automatically turns off; conversely, if the moisture levels fall below the threshold, the pump activates. Following this process, the Atmega controller transmits the gathered data to the LoRa transmitter.

On the receiving end, the node is equipped with components such as an LCD display, Wi-Fi module, Atmega 328 controller, and a LoRa receiver. The operational sequence initiates with the provision of power to the controller. The LoRa receiver captures signals transmitted by the transmitter and forwards them to the Atmega 328 controller for processing. Simultaneously, the received values are showcased on the LCD display, providing localized information. In parallel, the system is designed to upload the received parameters to the cloud server, specifically Thingspeak.com, ensuring that users can access real-time and historical greenhouse data conveniently. This comprehensive approach integrates both local and remote monitoring, allowing users to stay informed about the greenhouse conditions effortlessly.

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8. **Project prerequisite:** The following requirement list required for implementation for a monitoring system for greenhouse using WSN.

Sr.no.	Hardware Interface	Software Interface
1.	Temperature Sensor	Express PCB
2.	Humidity Sensor	Arduino UNO
3.	Light Sensor	Thing's Speak.com
4.	Air Quality Sensor	
5.	Moisture Sensor	
6.	LCD	
7.	Lora Module	
8.	Relay Driver	
9.	Relay	
10.	Water Pump	
11.	Atmega328 Controller	
12.	Esp32(Wi-Fi module)	
13.	Lora Module	

Table No.2: Hardware & Software project prerequisite

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9. **Hardware Specification**

1. LoRa Module

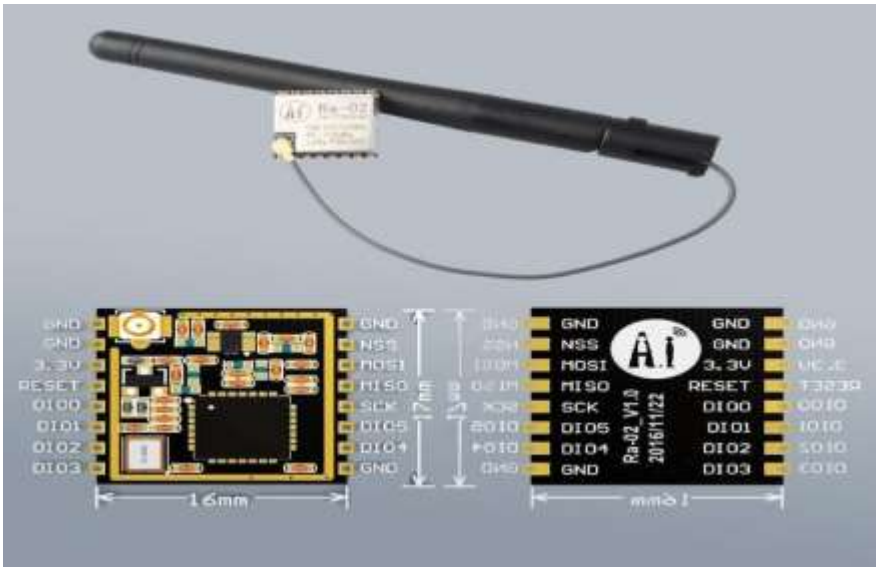

Figure No.3 Lora module

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LoRa spread spectrum modulation technology, Receive sensitivity as low as -141 dBm, excellent resistance to blocking, Supports preamble detection, and Supports half-duplex SPI communication Programmable bit rate up to 300Kbps.Supports FSK, GFSK, MSK, GMSK, LoRa and OOK modulation modes, Supports automatic RF signal detection. CAD mode and ultra-high speed AFC Packets with CRC, up to 256 bytes. Small package with double volume stamps

10. Results

Figure No.4 Working module of LoRa WAN

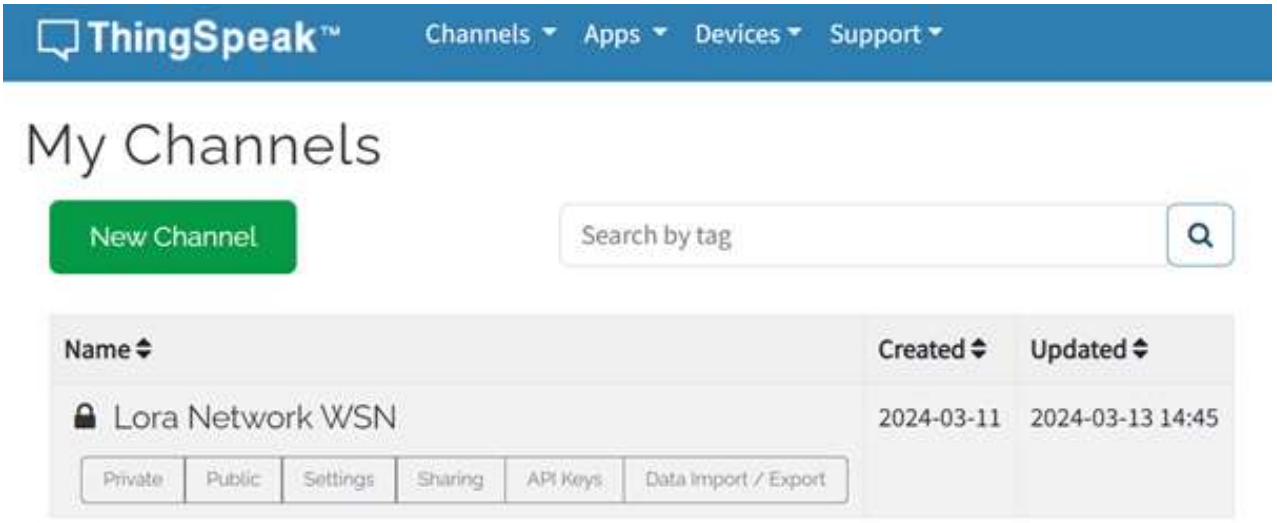

Figure No.5 Cloud Channel for Sensor values

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Figure No.6 Graphical Representation of Temperature sensor (Temperature over Time)

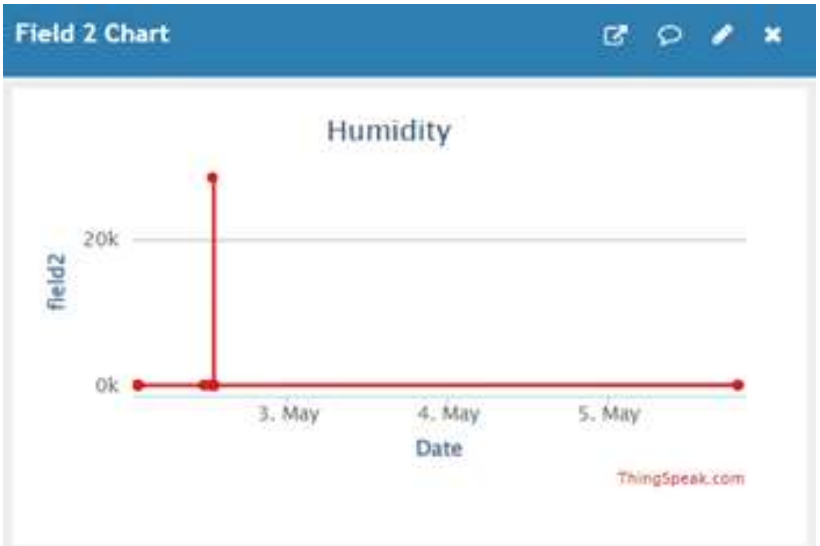

Figure No.7 Graphical Representation of Humidity sensor (Humidity over Time)

Figure No.8 Graphical Representation of Light sensor (Light over Time)

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Figure No.9 Graphical Representation of Moisture sensor (Moisture over Time)

Figure No.10 Graphical Representation of Air Quality sensor (Air Quality over Time)

11. Future Work and Application

• Sensor Networks: LoRa can be used to create a network of sensors within a greenhouse. These sensors can monitor various environmental parameters like temperature, humidity, soil moisture, light levels, and CO2 levels. This data can be collected and analyzed to optimize the growing conditions for plants.

• Energy Efficiency: LoRa can be used to monitor energy consumption within the greenhouse. This data can be used to optimize energy usage, for instance, by scheduling heating or cooling systems during off-peak hours.

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12. References:

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